Customer Shopping Behavior Analysis

An end-to-end data analysis project uncovering spending patterns, customer segments, and product preferences from **3,900 real transactions** — using Python for data preparation, PostgreSQL for structured analysis, and Power BI for interactive dashboards.

PYTHON MYSQL POWER BI

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Dataset at a Glance

The dataset contains **3,900 rows** and **18 columns**, covering customer demographics, purchase details, and shopping behavior. Only 37 values were missing — all in the Review Rating column.

Customer Demographics

Age, Gender, Location, and Subscription Status

Purchase Details

Item Purchased, Category, Purchase Amount, Season, Size, Color

Shopping Behavior

Discount Applied, Promo Code Used, Previous Purchases, Frequency, Review Rating, Shipping Type

3,900

Transactions

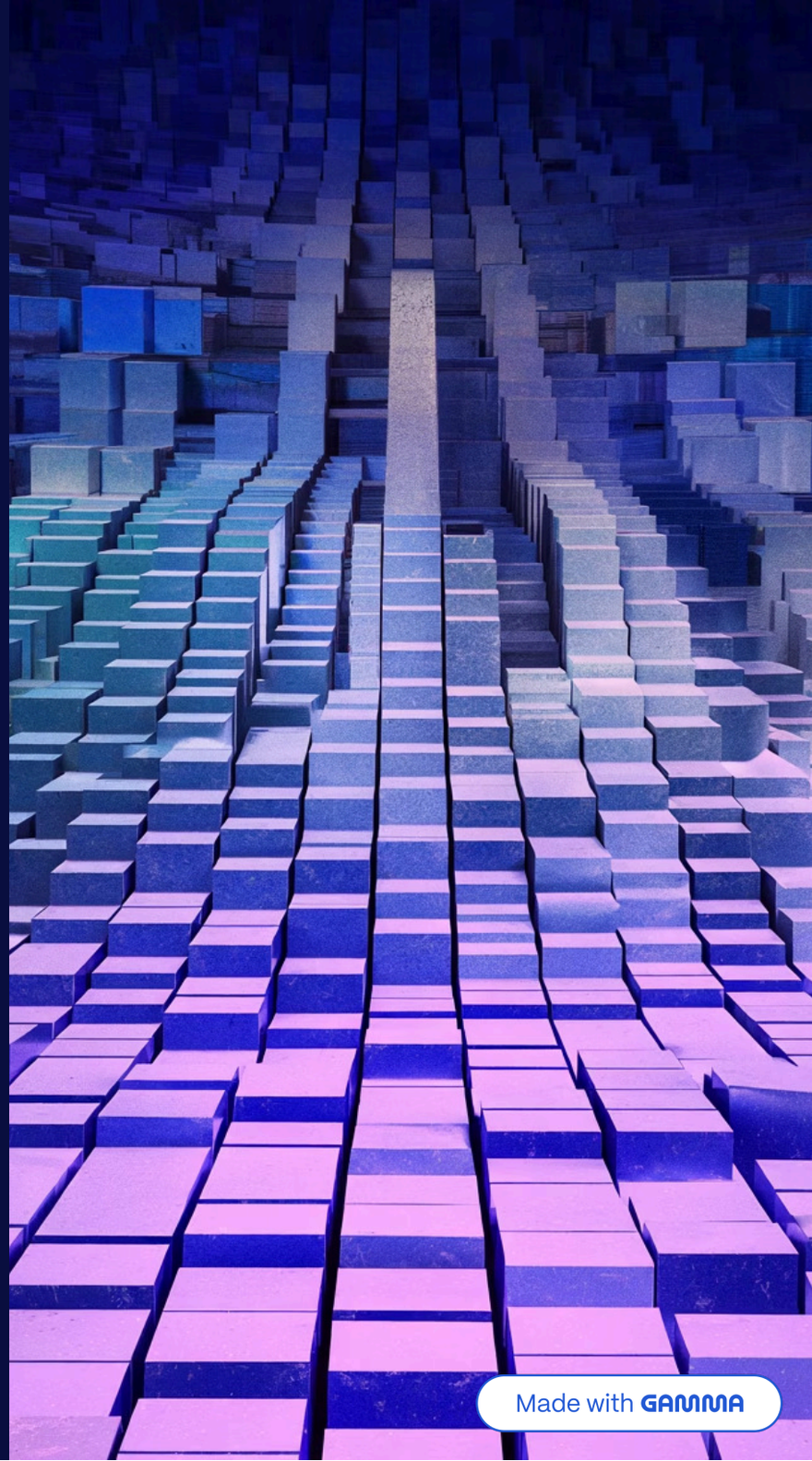
18

Columns

37

Missing Values

All in Review Rating



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Python: Data Preparation & Cleaning

Data Loading & Exploration

Imported the dataset using `pandas`. Used `df.info()` to check structure and `.describe()` for summary statistics across all 18 columns.

Missing Data Handling

Checked for null values and imputed missing `Review Rating` entries using the **median rating per product category**.

Column Standardization

Renamed all columns to **snake_case** for consistency and documentation.

Feature Engineering

- Created `age_group` by binning customer ages
- Created `purchase_frequency_days` from purchase data
- Verified `discount_applied` and `promo_code_used` were redundant — dropped `promo_code_used`

Database Integration

Connected Python to **PostgreSQL** and loaded the cleaned DataFrame for SQL analysis.

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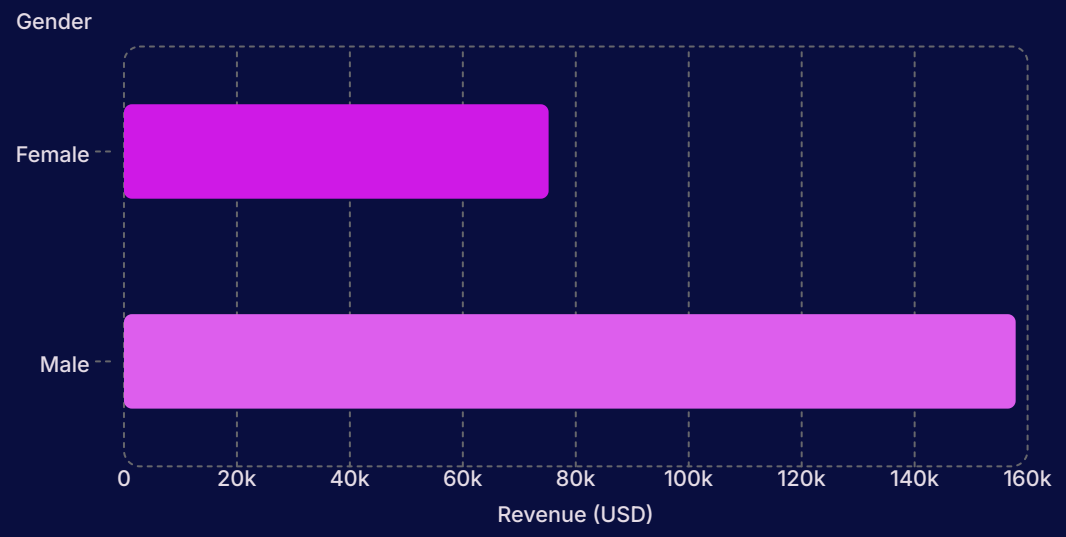
SQL Analysis: Revenue & Spending Patterns

Revenue by Gender



Male customers generated **$157,890** in revenue vs. **$75,191** for females — more than double.

High-Spending Discount Users

Identified **839 customers** who used discounts yet spent **above the average purchase amount** — indicating discount sensitivity among high-value buyers.

Shipping Type Comparison

Express shipping averaged **$60.48** vs. **$58.46** for Standard — a modest but consistent premium.

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SQL Analysis: Products & Ratings

Top 5 Products by Average Rating

Top 3 Products per Category

Category	Product	Orders
Accessories	Jewelry	171
Accessories	Sunglasses	161
Accessories	Belt	161
Clothing	Blouse	171
Clothing	Pants	171
Clothing	Shirt	169
Footwear	Sandals	160
Footwear	Shoes	150
Footwear	Sneakers	145
Outerwear	Jacket	163
Outerwear	Coat	161

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SQL Analysis: Customers & Subscriptions

Customer Segmentation

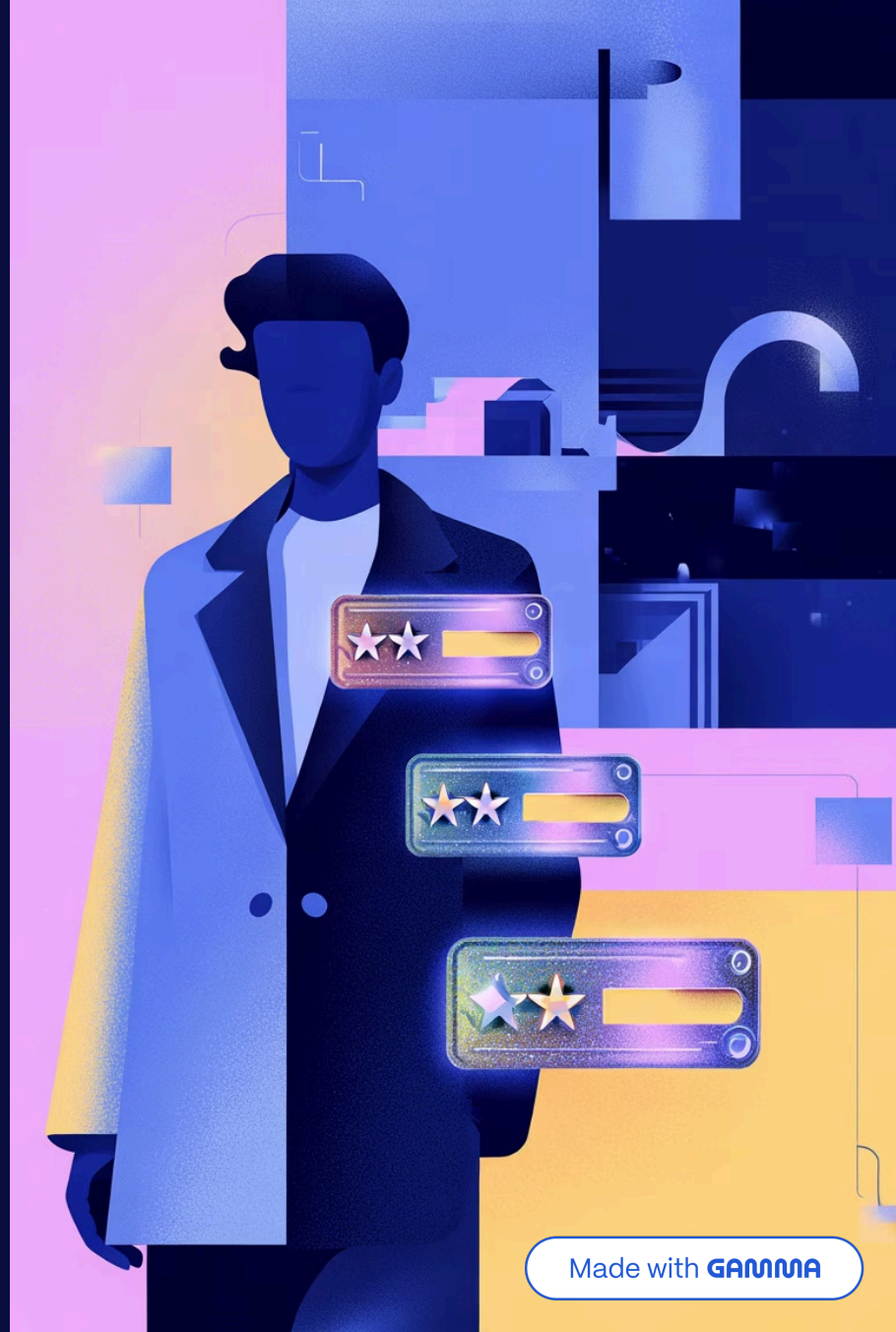
Classified customers by purchase history: **Loyal** (3,116), **Returning** (701), and **New** (83). The vast majority are loyal buyers.

Subscribers vs. Non-Subscribers

Non-subscribers (2,847) generated **$170,436** in revenue vs. subscribers (1,053) at **$62,645**. Average spend was nearly identical: $59.87 vs. $59.49.

Repeat Buyers & Subscriptions

Among customers with **>5 purchases**, 958 were subscribers and 2,518 were not — suggesting room to convert repeat buyers into subscribers.



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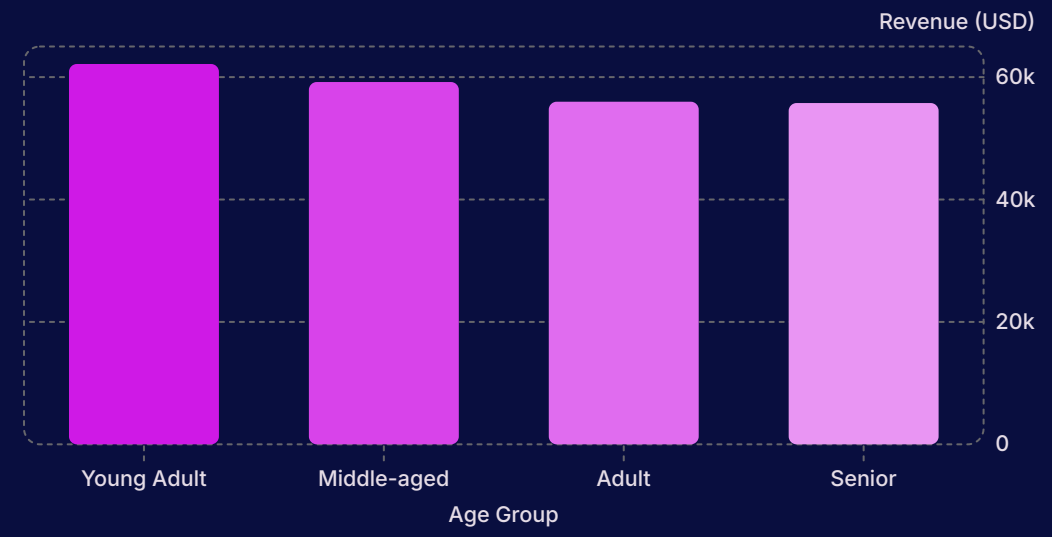
SQL Analysis: Discounts & Age Groups

Products Most Dependent on Discounts

These 5 products had the **highest percentage of discounted purchases**, with Hat at 50% — nearly half of all Hat sales involved a discount.

Revenue by Age Group



Young Adults led revenue at **$62,143**, with a gradual decline across older groups — all within a relatively tight range.

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Power BI Dashboard

An interactive **Power BI dashboard** was built to present all insights visually, with filters for Subscription Status, Gender, Category, and Shipping Type.

👥 3.9K	💳 $59.76	⭐ 3.75
Total Customers	Avg Purchase Amount	Avg Review Rating

Visualizations include: % of Customers by Subscription (27% Yes, 73% No), Revenue & Sales by Category, and Revenue & Sales by Age Group — all filterable for drill-down analysis.

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Key Findings Summary

Male Revenue Dominance

Male customers generated **$157,890** — more than double female revenue ($75,191) — a critical segment for targeting.

Loyal Customer Base

3,116 of 3,900 customers are classified as Loyal. Only 83 are New — retention is strong, but acquisition may need attention.

Discount Dependency

Top 5 discount-dependent products range from **47–50% discounted**. Hat leads at 50% — margin risk requires review.

Subscription Opportunity

Only **27% subscribe**, yet 958 repeat buyers (with >5 purchases) are already subscribers — a clear conversion opportunity.

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Business Recommendations

→ Boost Subscriptions

Promote exclusive subscriber benefits to convert the 2,518 non-subscribing repeat buyers and grow the 27% subscriber base.

→ Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment — 701 Returning customers represent an untapped loyalty opportunity.

→ Review Discount Policy

Balance sales boosts with margin control, especially for Hat, Sneakers, and Coat where ~50% of sales involve discounts.

→ Product Positioning

Highlight top-rated products (Gloves, Sandals, Boots) and best-sellers (Blouse, Jewelry, Jacket) in marketing campaigns.

→ Targeted Marketing

Focus efforts on high-revenue age groups — particularly Young Adults ($62,143) — and express-shipping users who spend more on average.

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